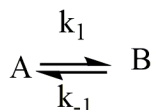


ACML101
Physical Chemistry Tutorial III

Instructor
Prof. Shashank Deep

Reversible Reaction

Q1. For an opposite reaction

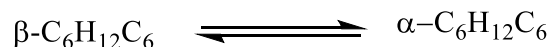


given $k_1 = 0.006 \text{ min}^{-1}$ and $k_{-1} = 0.002 \text{ min}^{-1}$, if only A exists at the beginning with a concentration of 1 mol.dm^{-3} , determine:

- (1) The time needed when concentrations of A and B are equal.
- (2) Concentrations of A and B after 100 min.

Ans: (1) 137 min (2) [A]=0.587 mol L⁻¹, [B]=0.413 mol L⁻¹

Q2. The reaction of β -glucose changing to α -glucose

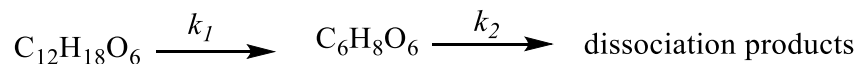


is a first order opposite reaction with $k_1 + k_{-1} = 0.0016 \text{ min}^{-1}$ and an equilibrium constant $K_c = 0.557$. Calculate k_1 and k_{-1} .

Ans: $k_1 = 0.0415 \text{ min}^{-1}$, $k_{-1} = 0.00745 \text{ min}^{-1}$

CONSECUTIVE REACTION AND SSA

Q3. The reaction of hydrolysis of L -2, 3 - ; 4, 6 - diacetone guluronic acid ($\text{C}_{12}\text{H}_{18}\text{O}_6$) in acid solution forming ascorbic acid ($\text{C}_6\text{H}_8\text{O}_6$, Vitamin C),

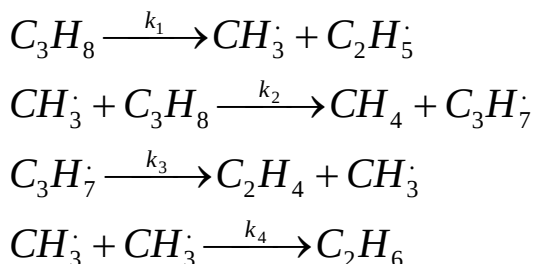


is a first order consecutive reaction with $k_1 = 4.2 \times 10^{-3} \text{ min}^{-1}$, $k_2 = 0.020 \times 10^{-3} \text{ min}^{-1}$ at 50°C and $k_1 = 410 \times 10^{-3} \text{ min}^{-1}$, $k_2 = 0.090 \times 10^{-3} \text{ min}^{-1}$ at 60°C . Determine the most

favourable reaction time and the corresponding maximum yield for the production of ascorbic acid.

Ans: At 50 C: 1279 min, 97.5%; At 60 C: 475 min, 95.8%

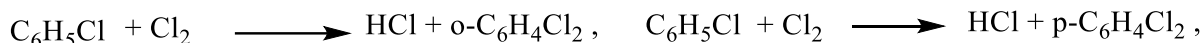
Q4. The following is a highly simplified mechanism for the decomposition of C_3H_8 :



Pick out the intermediates and formulate a steady state expression, where possible. From these, find expressions for each steady state concentration, and then formulate the overall rate of reaction in terms of the rate of production of CH_4 .

Parallel Reaction

Q5. For a parallel reaction of chlorobenzene and chlorine in a CS_2 solution,

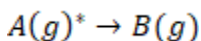
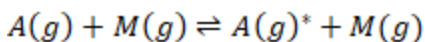


in the presence of iodine as a catalyst, with the same initial concentrations of 0.5 mol.dm^{-3} for C_6H_5Cl and Cl_2 at a constant temperature and a constant concentration of iodine, after 30 min, 15 % of C_6H_5Cl transforms to $o-C_6H_4Cl_2$, 25 % to $p-C_6H_4Cl_2$. Calculate the second order rate coefficients k_1 and k_2 for the two reactions.

Ans: $k_1 = 0.166 \text{ L mol}^{-1} \text{ min}^{-1}$, $k_2 = 0.0278 \text{ L mol}^{-1} \text{ min}^{-1}$

Temperature dependence of the rate constant

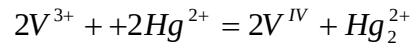
(6) Consider a reaction $A(g) \rightarrow B(g)$ taking place in two elementary steps in presence of a catalyst M



Calculate k_{obs} in terms of rate constants (k_1 , k_{-1} and k_2). If $k_{-1}[M] \gg k_2$, calculate observed activation energy in terms activation energy (threshold energy) of individual steps.

Mechanism of the reaction

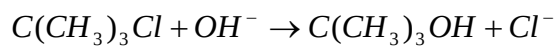
Q7. Suggest one reaction scheme for the oxidation of vanadium (III) ions by mercury (II) ions on the basis of rate equation given. Please provide the logic behind the proposed mechanism.



$$-\frac{d[V^{3+}]}{dt} = \frac{k[V^{3+}]^2[Hg^{2+}]}{k'[V^{IV}] + [V^{3+}]}$$

Mechanism of reaction from the rate law

Q10. Suggest plausible mechanisms for the following reaction:



$$\frac{-d[C(CH_3)_3Cl]}{dt} = \frac{k[C(CH_3)_3Cl][OH^-]}{k'[Cl^-] + k''[OH^-]}$$